0. **Code:** Tanuboddy Test3.zip

1. ConsoleApp2 / Program.cs
2. ConsoleApp3 / Program.cs
3. ConsoleApp4 / Program.cs
4. ConsoleApp5 / Program.cs
5. ConsoleApp6 / Program.cs
6. ConsoleApp7 / Program.cs

1. **ConsoleApp2 / Program.cs**

```
using System.Buffers.Text;

namespace Base_n
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter an integer");
            int a = Convert.ToInt32(Console.ReadLine());
            if (a < 0)
            {
                a = -a;
            }

            Console.WriteLine("Enter a base-N");
            int b = Convert.ToInt32(Console.ReadLine());
            while (b ≤ 0)
            {
                Console.WriteLine("Enter a base-n");
                b = Convert.ToInt32(Console.ReadLine());

                if (b > 0)
                    break;
                else
                    Console.WriteLine("Enter a base-n");

            }

            int mod;
            int x = a % b;
```

A1

A3

A2

1. **ConsoleApp2 / Program.cs**

```
            int large;
            Console.WriteLine("Modul is " + x);
            large = x;
            int sum = 0;
            sum = sum + x;
            int y = a / 10;
            Console.WriteLine("Updated sum is " + sum);
            Console.WriteLine("Updated integer is " + y);
            while (y > 0)
            {
                mod = y % b;
                Console.WriteLine("Modul is " + mod);
                if (mod > large)
                {
                    large = mod;
                }
                sum = mod + sum;
                Console.WriteLine("Updated sum is " + sum);
                y = y / 10;
                Console.WriteLine("Updated integer is " + y);
            }
            Console.WriteLine("Sum of all Base-N digits is " + sum);
            Console.WriteLine("The largest base-n digit is " + large);

        }
    }
}
```

first A5

bi

A3

A5

A4

bi

2. ConsoleApp3 / Program.cs

```
namespace ConsoleApp3
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter an integer");
            int a = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine("Enter a base-N");
            int b = Convert.ToInt32(Console.ReadLine());

            int mod = a % b;
            while (true)
                int x = a / b;

            Console.WriteLine(x);
            Console.WriteLine(mod);


        }
    }
}
```

3. ConsoleApp4 / Program.cs

```
namespace ConsoleApp4
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter a number");
            int a = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter base 10");
            int b = Convert.ToInt32(Console.ReadLine());

            Console.WriteLine(a % b);
        }
    }
}
```

4. ConsoleApp5 / Program.cs

```
namespace InsertionCounts
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] arr = {
                23,
                1,
                10,
                5,
                2
            };
            int[] count = CountInsertions(arr);
            Console.WriteLine("Count array:");
            foreach(var num in count)
            {
                Console.WriteLine(num + " ");
            }
        }
        static int[] CountInsertions(int[] arr)
        {
            int n = arr.Length;
            int[] count = new int[n];
            for (int i = 1; i < n; i++)
            {
                int key = arr[i];
                int j = i - 1;
                int insertioncount = 0;
                while (j ≥ 0 && arr[j] == key)
                {
```

B1

B2

arr[j] > key

B3

4. **ConsoleApp5 / Program.cs**

```
                    arr[j + 1] = arr[j];
                    j--;
                    insertioncount++;
                }
                arr[j + 1] = key;
                count[i] = insertioncount;
            }
            return count;
        }
    }
}
```

good.

B4 Sorting an array.

5. ConsoleApp6 / Program.cs

```
namespace dictionary
{
    internal class Program
    {
        static void Main(string[] args)
        {
            string input = Console.ReadLine();
            int[] arr = ParseInput(input);
            int result = SumofLargenumbers(arr);
        }
    }
}
```

6. ConsoleApp7 / Program.cs

```
using System;

namespace ConsoleApp7
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] array = {
                1,
                2,
                3,
                2,
                7,
                3,
                4,
                4,
                5,
                4,
                5,
                1,
                6,
                1,
                1
            };
            var occurences = new Dictionary < int,
                int > ();
            var maxFrequency = new Dictionary < int,
                int > ();
            foreach(int num in array)
            {
```

6. **ConsoleApp7 / Program.cs**

```
            if (occurences.ContainsKey(num))
            {
                occurences[num]++;
            }
            else
            {
                occurences[num] = 1;
            }
            foreach(var pair in occurences)
            {
                int freq = pair.Value;
                int numb = pair.Key;
                if (numb > maxFrequency[freq])
                {
                    maxFrequency[freq] = numb;
                }
                else
                {
                    maxFrequency[freq] = freq;
                }
            }
        }
    }
}
```

C2

Separate loop.

needs to check for contains key